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Instructions

- Complete your work in this Word document.
 - Write your R code in RStudio, take screenshots of both your code AND the output/plots, and insert them into this document.
 - Ensure all screenshots are clear and readable.
 - Include all written answers and interpretations in complete sentences.
 - Convert your final document to PDF before submission.
 - **Submit only ONE PDF file.** Do not submit .docx or .R files.
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Before you begin: **set.seed(42)**

Scenario

A cloud monitoring team tracks the response times of an internal microservice. Based on historical logs, the population of response times follows a normal distribution with:

- Population mean: $\mu = 250$ ms
- Population standard deviation: $\sigma = 40$ ms

The team wants to understand how **sample means** behave when they randomly sample batches of requests to estimate average response time.

a) (2 marks) Write R code to simulate drawing 1000 random samples of size $n = 25$ from this population. For each sample, calculate the sample mean and store all 1000 means in a vector.

Write your R code below and insert a screenshot of your code and console output confirming it ran successfully.

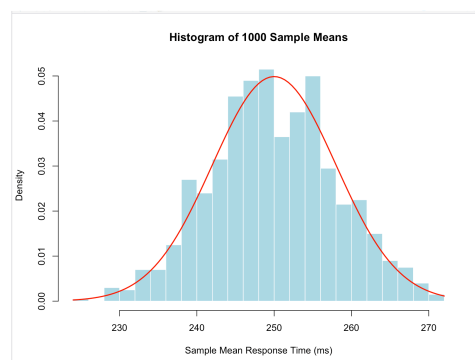
```
# part a
set.seed(42)
n= 25
mean = 250
sd = 40
sample_means= replicate(1000, mean(rnorm(25, mean = 250, sd = 40)))
```

```
> set.seed(42)
> n= 25
> mean = 250
> sd = 40
> sample_means= replicate(1000, mean(rnorm(25, mean = 250, sd = 40)))
> |
```

b) (3 marks) Create a histogram of the 1000 sample means and overlay the theoretical sampling distribution curve. Label your axes clearly.

```
# part b
hist(sample_means, breaks = 30,
     main = "Histogram of 1000 Sample Means",
     xlab = "Sample Mean Response Time (ms)",
     ylab = "Density",
     col = "lightblue", border = "white",
     probability = TRUE)

curve(dnorm(x, mean = 250, sd = 40/sqrt(25)),
     add = TRUE, col = "red", lwd = 2)
```



c) (3 marks) By using the simulated data, calculate the mean and standard deviation of your 1000 sample means. What are the theoretical values for mean and standard deviation of the sample means? How do the experimental values compare to the theoretical values?

```

> # part c
> mean(sample_means)
[1] 249.9375
> sd(sample_means)
[1] 8.147651
>

```

Theoretical values:

Mean = μ = **250 ms**

sd(Standard Error) = $\sigma/\sqrt{n} = 40/\sqrt{25} = \mathbf{8\ ms}$

The experimental values will be very close to mean ≈ 250 and SD ≈ 8), confirming the CTL

Experimental mean = **249.9375** ≈ 250

Experimental SD = **8.147651** ≈ 8

d) (4 marks) What percentage of your sample means fall within 1.2 standard error of the population mean? Solve both theoretically and experimentally, then compare

experimental result should be very close to **76.99%**, confirming the theoretical value.

```

# part d Theoretical
(pnorm(1.2) - pnorm(-1.2)) * 100

```

```

> # part d Theoretical
> (pnorm(1.2) - pnorm(-1.2)) * 100
[1] 76.98607
>

```

```

# part d experimental
se = 40 / sqrt(25)
mean(sample_means >= 250 - 1.2*se & sample_means <= 250 + 1.2*se) * 100

```

```

> # part d experimental
> se = 40 / sqrt(25)
> mean(sample_means >= 250 - 1.2*se & sample_means <= 250 + 1.2*se) * 100
[1] 74.4
>

```